(Please write your Enrollment Number) Enrollment No. 01601172021

End-TermExamination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech,CSE-AI, Semester:<VI>
(May, 2024)

Subject Code: BAI 302	Subject: Natural language processing
Time :3 Hours	Maximum Marks :60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		~~(2.5*8=20)~~ (2.5*2)+(3*5)	
	(a) What is stop words and library used for stop words in NLP?	2.5	
	(b) Explain any two Application and challenge in NLP.		
	(c) Explain Chat bots give two names of popular chatbots.	3 marks each	
	(d) Explain NLTK, Spacy?		
	(e) Comparison of Word-based and phrase- based machine.		
	(f) What is Transformer model?		
	(g) Discuss the advantage and disadvantage of LSTM?		
	UNIT-I		
Q2	Elaborate on Natural Language Processing (NLP) and its pipeline, including a diagram illustrating the process, along with examples for each step?	(10)	
Q3	For the sentence "They are hosting a cooking class where participants learn essential culinary techniques. Mastering these skills leads to the creation of delicious meals, enhancing one's culinary prowess and creativity in the kitchen." Perform all type of tokenization and stop words remove.	(10)	
	UNIT-II		
Q4	Create Word2vec vectors using CBOW Model and one- hot encodes vector representation. S1: this is a makeup exam. S2: exam is on fairness. S3: fairness makes machines better.	(10)	
Q5	Create TF-IDF vectors using Term-document matrix. S1: They is a leadership seminar. S2: Seminar focuses on communication skills. S3. Communication Skills enhance workplace productivity.	(10)	
	UNIT-III		
Q6	Explain Rule-based Machine Translation and Neural Machine Translation, incorporating diagrams, examples, and discussions on their respective challenges. Additionally, provide a comparison highlighting the distinctions between the two approaches.	(10)	
Q7	Explain Sequence to sequence model and Attention model, incorporating diagrams, examples, and discussions on their respective challenges. Additionally, provide a comparison highlighting the distinctions between the two approaches.	(10)	
	UNIT-IV		
Q8	Differentiate between RNN and CNN explain each in detail with diagram.	(10)	
Q9	Explain Gated Recurrent Unit (GRU) architecture and explain challenges, application, and merits with diagram.	(10)	